

Project plan

IoT Measurement Setup

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Approach and Planning

Approach

Throughout the project, the scrum methodology is used. Meaning that there are made use of sprints to keep track of the process and the next steps. Each sprint lasts 3 weeks, with exceptions due to holidays. Within a sprint multiple meeting types are used to ensure for good communication and flow. The first meeting starts at the the first day of the sprint. During this meeting there is looked back out the previous sprint, the review and retrospective, and the planning for the new sprint is made. At the end of a sprint, a sprint demo takes place. Meaning that there is showcased what has been done and achieved. Lastly, on a daily basis short stand ups take place. These stand ups are used to ensure that everyone is on tracking with what they are doing and if any help is necessary.

Test Approach

In order to allow for a flawless software implementation, testing is used. Testing is done in different manners with each manner testing a different aspect. The main forms of testing that will be used during this project is any form of unit tests and integration tests.

Breakdown of the Project

The project is divided into 4 different phases, each phase serves its purpose. The phases are created to serve as a clear guidance throughout the project. Next to this, it also ensures the project stays on track and clear overview of what is going on.

Analysis and problem definition

The first phase is the most important as it serves to create a solid foundation for the project. During this phase the focus lies on defining the problem, finding solutions and creating a concrete plan. The plan is used during the entirety of the project to ensure the final project aligns with the client's wishes.

Design

During the second phase, the focus is shifted to creating the design of the system. This means that the overall architecture of the system is shaped with the help of different diagrams, such as flow charts and state diagrams.

Realization

During second last phase, the implementation of the plan and design is made into reality. All the gained knowledge is challenged and put into practice. In a continuous cycle the software is implemented and tested. Once the planned result is reached, the next phase starts.

Evaluation

The last phase is focused on showcasing all that is learned and created during the previous phases. Meaning that a demonstration of the final product is given, and evaluated.

Timeline

There are multiple timelines that will guide the course of the project. The first timeline is the overall timeline, which gives the basic structure of the project, when the sprint are, etc. The second timeline is focussed on the tasks that are preformed during each sprint.

Overall Timeline

The timeline given in [\[Project Timeline\]](#) shows the overall overview timeline of the project, and when some important events take place.

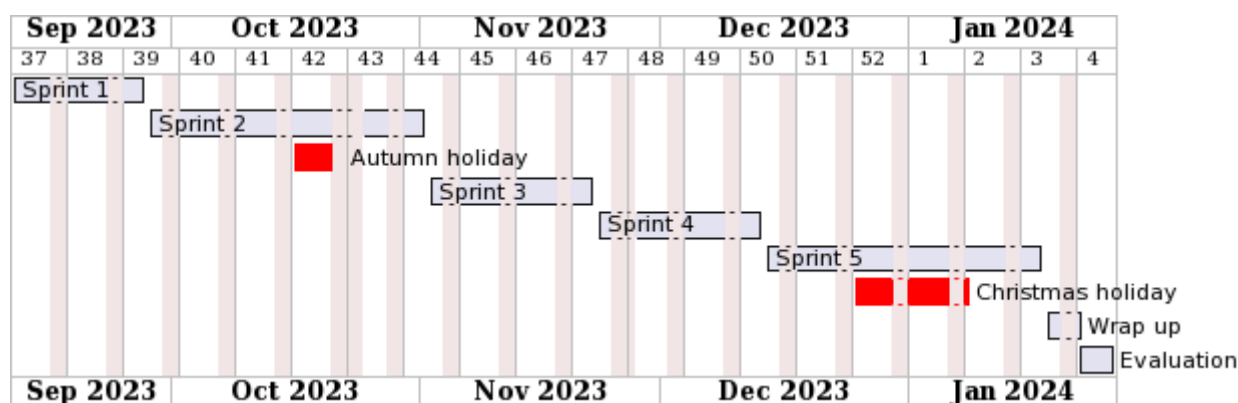


Figure 1. Project Timeline

Tasks Timeline

The phases, and the associated tasks, are given in a table to give an clear overview of how the project is approach and estimation of phasing realization. The table [\[Expected Tasks and phasing duration\]](#) is more detailed and will give 'more' detail on the expected tasks that will be preformed during the sprint. In the timeline [\[Phasing Timeline\]](#) a more sketched out overview of the phasing throughout the weeks and sprints are given.

Table 1. Expected Tasks and phasing duration

Sprint	Phase	Tasks
1	Analysis and Problem Definition	Research the problem, the solutions, and its requirements. Next to this, create the concrete plan.
2	Design	Create the design.
3	Realization	Implementation and testing.
4	Realization	Implementation and testing.
5	Realization / Evaluation	Implementation, testing and final realization.

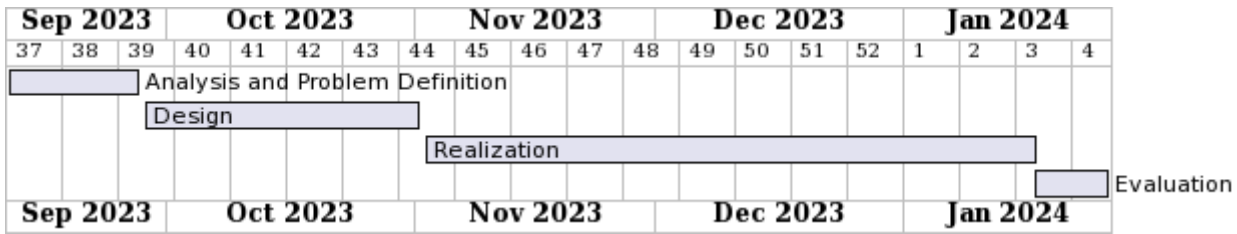


Figure 2. Phasing Timeline

Finance and Risks

Finance

While the primary objective of this project is centered around the development of the prototype, it's essential to acknowledge the potential need for specific hardware components that may not be available at the ISSD department. To address this potential requirement and ensure the project's successful execution, a financial plan has been established.

Hardware Acquirement

In the event that specific hardware is necessary for the project, the project team will assess this with the HTES during the planning phase. This assessment will involve identifying any hardware components or equipment that cannot be sourced internally within the ISSD. These could include unique sensors, microcontrollers, or other specialized devices essential for the prototype's functionality.

Collaboration with HTES

To cover the financial costs for any of the non-available hardware, the project team will approach the HTES for funding, as they are the client. Upon approaching the HTES, the necessary hardware is discussed to validate the purchase. Based on the result of the discussion, the funding is allocated or a different solution has to be found.

Risks and Fallback Activities

With every project risks are involved. Each risks with their own impact and solution. To display the risks we could potentially could run into, how we could solve them and ultimately prevent them, see [\[Risk and Fallback Activities\]](#).

Table 2. Risk and Fallback Activities

Risk	Prevention activities included in the plan	Fall-back Activities
Running out of time.	Make a concrete plan with margins. Planning a bit more time for each phase can help to get behind when problems do occur.	Discuss with coaches where to go from now.

Risk	Prevention activities included in the plan	Fall-back Activities
Absence of a team member.	Have clear communication on the current situation. And if the absences of a team member take too long, direct action can be taken.	Contact a different team member for the needed help.
Miscommunication.	Be sure to attend the meetings and communicate the current progress and what the plans are. If something doesn't line up, this can be discussed right away.	Make use of the different communication channels, and if necessary, meetings to get back on the same page. Besides that, the GitLab version control can be used to revert any changes that were made but not wished for.
Hardware issues.	Plan ahead making sure that everything will be there on time and if issues do occur contact the right person right away.	Communicate the problems with the team and find a solution on how it can be solved. In the meantime, documentation can be picked up.
Late change in requirements.	Keep in close touch with the project lead, to keep track of what's going on and to get an hint of changes.	Discuss what is feasible and/or of the change in requirements is truly necessary.
Software not up to their standards.	Have frequent code reviews to keep track of how your code fits into their standards.	Discuss the situation and find possible solutions on how to make it up to their standards.
Unplanned additions or work that are added to the work load.	Keep in close touch with project lead, to keep track of how the project is coming along and the wiggle room of additions is.	Come to an altermatum with the project lead to make sure both parties do not suffer any consequences.